

ARTIFICIAL INTELLIGENCE: CONCEPTS AND TECHNIQUES

WEEK 12:

Lecture 55

This document, "lecture55.pdf," provides a comprehensive overview of Natural Language Processing (NLP). Here's a summary of its key sections:

The document begins by defining NLP as a field in computer science, AI, and computational linguistics concerned with interactions between computers and human languages. It emphasizes giving computers the ability to understand and manipulate text and spoken words like humans do.

It then outlines what can be achieved with NLP, including:

- Talking to phones.
- Using the web to answer questions.
- Mapping out discussions.
- Translating between languages.
- Summarizing documents.
- Question answering.

The document also lists areas related to NLP, such as information retrieval, machine-readable dictionaries, speech synthesis and recognition, and cross-language information retrieval.

Major NLP tasks are detailed across several pages, encompassing:

- Automatic summarization and abstracting.
- Coreference resolution.
- Discourse analysis.
- Machine translation.
- Morphological analysis.
- Named entity recognition (NER).
- Optical Character Recognition (OCR).
- Part-of-speech (POS) tagging.
- Parsing.
- Relationship extraction.
- Sentence breaking.
- Sentiment analysis.
- Speech recognition and segmentation.
- Topic segmentation and recognition.
- Word segmentation and word sense disambiguation.
- Information retrieval and extraction.

Other tasks mentioned include native language identification, stemming, speech simplification, text-proofing, natural language search, query expansion, automated essay scoring, and truecasing.

The concept of natural language communication is discussed, highlighting sentences, phrase structures, and different types of grammars (recursive enumerable, context-sensitive, context-free, and regular). It also presents examples of grammar for a natural language.

Ambiguities in language are explored, covering lexical, syntactic, and semantic ambiguity, as well as the ambiguity between literal and figurative meaning (metonymy and metaphor).

Disambiguation is explained as the process of recovering the most probable intended meaning, involving world, mental, language, and acoustic models.

Communication between agents is described, focusing on speech acts like informing, querying, answering, requesting, promising, and acknowledging. The ASK and TELL functions for agent communication are introduced, along with associated difficulties such as naming policies and reconciling knowledge base differences.

The component steps of communication are broken down into seven processes: intention, generation, synthesis (speaker's side), and perception, analysis, disambiguation, and incorporation (hearer's side).

The document delves into representing documents, mentioning the vector space model and the Bag of Words (BoW) model as orderless document representations that keep word frequencies.

Turning text into features is covered, with an emphasis on word embeddings. Techniques like integer representation, one-hot encoding, bag-of-words, and TF-IDF are explained. Popular word embedding techniques such as Word2vec, GloVe, and fastText are introduced.

Natural Language Understanding (NLU) is presented as a multi-level process starting from word-level, moving to sentence-level, and then to context and overall environment. It discusses POS tagging, parsing, lexical semantics, compositional semantics, and Word Sense Disambiguation (WSD).

Text analytics is defined as the process of drawing meaning from written communication, using text mining and NLP algorithms to uncover patterns in unstructured data. Applications include sentiment analysis, email spam filters, automated ad placement, social media monitoring, enterprise business intelligence, national security, scientific discovery, and competitive intelligence.

Computational Linguistics is presented as the field that creates tools for practical tasks like machine translation, speech recognition, and information extraction.

Named Entity Recognition (NER) is explained as identifying and categorizing specific words or phrases (e.g., persons, locations, organizations, dates, measures). Three standard approaches to NER are listed: handwritten regular expressions, classifiers, and sequence models.

Various NLP software suites and libraries are mentioned, including Stanford CoreNLP, NLTK, spaCy, Apache OpenNLP, DeepLearning4J, and Hugging Face.

Recent research in NLP focuses on Natural Language Generation (creative writing, news generation) and Natural Language Understanding (sentiment analysis, text

summarization, information extraction, topic analysis, question answering, parsing, POS tagging, translation).

Important NLP models are introduced: Eliza, Tay, BERT, GPT-3, LaMDA, and Mixture of Experts (MoE).

Finally, the document concludes with the benefits and limitations of NLP, including difficulties with sarcasm, emotion, slang, context, and high computational requirements.

Lecture 56

This document, "Lecture 56: Ethics," defines AI ethics as the guiding principles for the responsible development and deployment of AI, focusing on human values and societal well-being.

The lecture emphasizes why AI ethics matters:

- **Avoiding Harm:** Preventing AI from causing harm to individuals or society.
- **Building Trust:** Fostering trust in AI technology and its developers.
- **Ensuring Responsible Innovation:** Encouraging developers to consider the potential impact of their work.
- **Promoting Inclusivity:** Ensuring AI systems benefit all members of society.
- **Preventing Bias:** Mitigating harmful consequences from biased or inaccurate data.

Key aspects of AI ethics include fairness, transparency, accountability, privacy, security, human-centricity, environmental sustainability, inclusion, value alignment, and trust.

The document discusses ethical challenges such as:

- AI and bias
- AI and privacy
- AI and the environment

The Belmont Report offers three principles for experiment and algorithm design:

1. **Respect for Persons**
2. **Beneficence**
3. **Justice**

AI ethics in practice involves:

- AI Impact Assessments
- AI Ethics Committees
- AI Codes of Ethics
- AI Regulations

Various stakeholders contribute to AI ethics, including academics, governments, intergovernmental entities (UN, World Bank), non-profit organizations, and private companies (IBM, Google, Meta).

Organizations promoting AI ethics include AlgorithmWatch, AI Now Institute, DARPA, CHAI, and the National Security Commission on Artificial Intelligence.

IBM's view on AI ethics highlights:

1. AI should augment, not replace, human intelligence.
2. Data and insights belong to their creator.
3. AI systems must be transparent and explainable.

The document also mentions Isaac Asimov's Three Laws of Robotics.

Lecture 57

This document, "Lecture 57: Swarm Intelligence (SI)," provides a comprehensive overview of Swarm Intelligence, its underlying principles, various algorithms inspired by it, and detailed explanations of two prominent SI models: Ant Colony Optimization (ACO) and Particle Swarm Optimization (PSO).

Key sections include:

1. Definition and Biological Inspiration
2. Swarm Inspired Algorithms
3. Swarm Behavior of Different Animals/Birds/Insects
4. Principles Used in SI
5. Emergence
6. General Framework
7. Advantages of SI
8. Limitations of SI

Detailed models:

- **Ant Colony Optimization (ACO Model):**

Inspired by the social behavior of ant colonies, specifically their ability to find shortest routes to food. Ants communicate via pheromones, and artificial ants use this concept to solve optimization problems.

- **Particle Swarm Optimization (PSO):**

Introduced by Eberhart and Kennedy (1995), inspired by bird flocking. It is used for non-linear continuous optimization problems. Particles adjust their positions based on personal best and global best solutions.

The lecture also explains similarities between PSO and Genetic Algorithms (GA), and hybrid approaches such as PSO-GA.

Lecture 58

This lecture on Multi-agent Systems (MAS) defines them as systems where multiple agents interact, often with different goals, requiring cooperation, coordination, and negotiation.

Key concepts and applications discussed include:

- Coordination

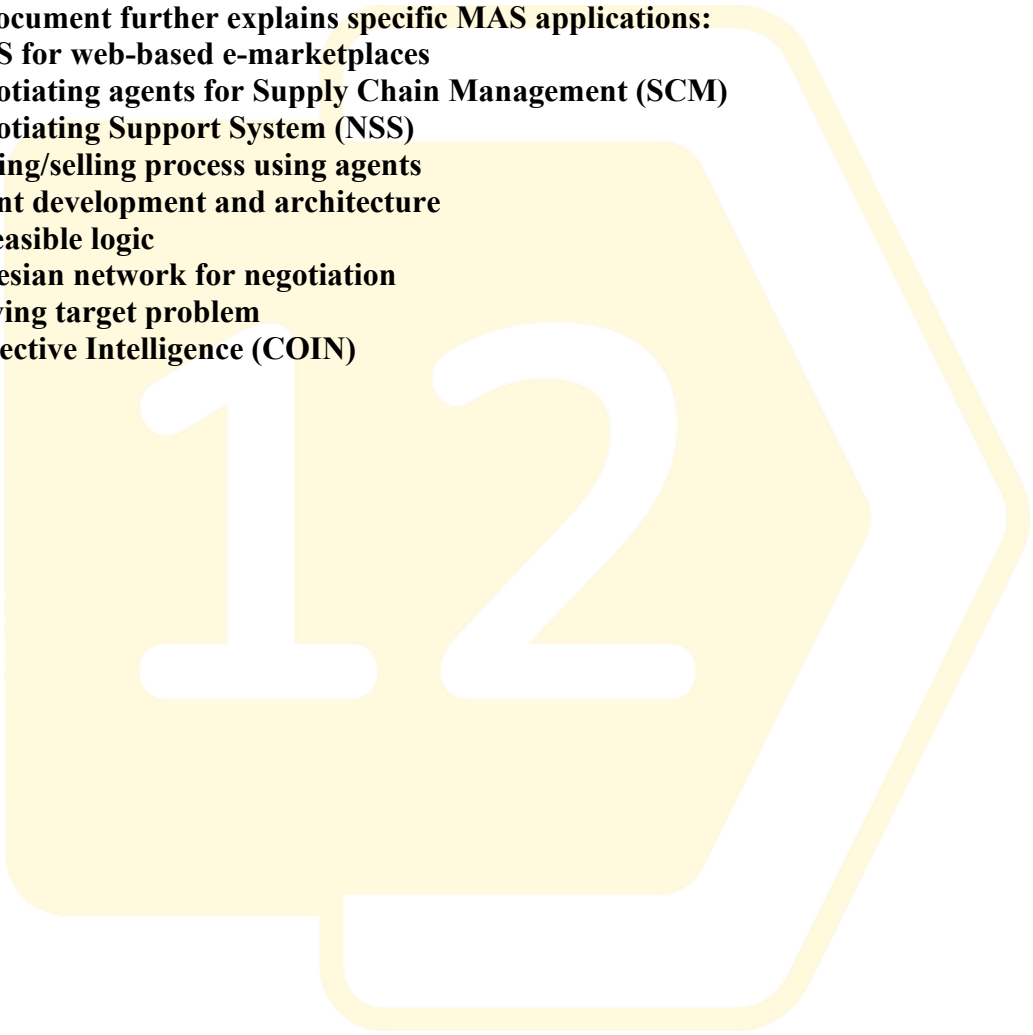
- **Cooperation**
- **Negotiation**

Real-world applications:

- **Graphics in computer games**
- **Internet**
- **Agent-Oriented Software Engineering**
- **Air traffic control**
- **Decision making**
- **Robotics**
- **Spacecraft control**

The document further explains specific MAS applications:

- **MAS for web-based e-marketplaces**
- **Negotiating agents for Supply Chain Management (SCM)**
- **Negotiating Support System (NSS)**
- **Buying/selling process using agents**
- **Agent development and architecture**
- **Defeasible logic**
- **Bayesian network for negotiation**
- **Moving target problem**
- **Collective Intelligence (COIN)**



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